

```

import { useState } from "react";

const layers = [
  {
    id: "regulatory",
    code: "L1",
    label: "Regulatory Foundation",
    color: "#C8A84B",
    icon: "🏛️",
    subtitle: "The Legal Perimeter",
    description: "Map every applicable obligation before a single line of tech is
    nonAI: [
      {
        name: "FinCEN BSA Program",
        detail: "Written AML policy, designated BSA Officer, independent audit cy
        action: "Draft & register",
        priority: "MANDATORY"
      },
      {
        name: "OFAC SDN & Consolidated Sanctions Lists",
        detail: "Subscribe to OFAC's daily-updated XML feeds. All wallets, counte
        action: "Live feed integration",
        priority: "MANDATORY"
      },
      {
        name: "FATF Travel Rule (Rule 31 CFR 1010.410)",
        detail: "Originator and beneficiary information must travel with every vi
        action: "Protocol adoption",
        priority: "MANDATORY"
      },
      {
        name: "State MSB Licensing Matrix",
        detail: "Map all 50-state licensing requirements. Nevada, California, and
        action: "Licensing tracker",
        priority: "MANDATORY"
      },
      {
        name: "CTR / SAR Filing Thresholds",
        detail: "CTR: cash transactions ≥ $10,000. SAR: suspicious activity ≥ $5,
        action: "Threshold engine",
        priority: "MANDATORY"
      }
    ],
    ai: [

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    {
      name: "Regulatory Change Intelligence",
      detail: "NLP agents continuously monitor FinCEN, OFAC, FATF, and SEC rule
      tech: "LLM + RAG pipeline",
      model: "GPT-4o / Claude"
    },
    {
      name: "Policy Q&A Assistant",
      detail: "Internal chatbot trained on your AML policy, FinCEN guidance, an
      tech: "Fine-tuned LLM",
      model: "Claude Haiku"
    }
  ]
},
{
  id: "kyc",
  code: "L2",
  label: "KYC / CDD / EDD",
  color: "#4B8FC8",
  icon: "🪪",
  subtitle: "Know Your Customer Stack",
  description: "The onboarding gate. Risk-tier every customer at intake. CDD fo
  nonAI: [
    {
      name: "Identity Document Verification",
      detail: "Collect government-issued ID (passport, DL, national ID). Verify
      action: "SDK integration",
      priority: "HIGH"
    },
    {
      name: "Liveness Detection",
      detail: "Biometric selfie match against ID photo. 3D liveness prevents sp
      action: "Biometric check",
      priority: "HIGH"
    },
    {
      name: "Beneficial Ownership (CDD Rule)",
      detail: "For legal entities: identify all individuals owning ≥25% equity
      action: "UBO form + verification",
      priority: "MANDATORY"
    },
    {
      name: "PEP & Adverse Media Screening",
      detail: "Screen all customers and UBOs against Politically Exposed Person
      action: "API screening",
      priority: "HIGH"
    },
  ],

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    {
      name: "Wallet Address Due Diligence",
      detail: "For crypto customers: verify self-hosted wallet ownership via mi
      action: "Proof-of-ownership flow",
      priority: "HIGH"
    }
  ],
  ai: [
    {
      name: "AI Document Intelligence",
      detail: "OCR + vision models extract and validate all fields from ID docu
      tech: "Computer Vision + OCR",
      model: "AWS Textract / Google Vision"
    },
    {
      name: "Dynamic Risk Scoring Engine",
      detail: "ML model assigns a 0-100 risk score at onboarding using 50+ feat
      tech: "Gradient Boosting / XGBoost",
      model: "Custom trained"
    },
    {
      name: "NLP Adverse Media Mining",
      detail: "Continuously crawl global news in 40+ languages. Entity recognit
      tech: "NER + Sentiment NLP",
      model: "spaCy + LLM reranker"
    }
  ]
},
{
  id: "txn",
  code: "L3",
  label: "Transaction Monitoring",
  color: "#C84B4B",
  icon: "🔍",
  subtitle: "Real-Time & Retrospective Surveillance",
  description: "The operational core. Every on-chain and off-chain transaction
  nonAI: [
    {
      name: "Blockchain Analytics Platform",
      detail: "Integrate Chainalysis Reactor, Elliptic, or TRM Labs. These map
      action: "API + webhook",
      priority: "MANDATORY"
    },
    {
      name: "Rule-Based Alert Engine",
      detail: "Deterministic rules: structuring detection (sub-$10K deposits in
      action: "Rules YAML config",

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    priority: "MANDATORY"
  },
  {
    name: "OFAC Real-Time Screening",
    detail: "Every transaction origin/destination address and counterparty na
    action: "Pre-settlement gate",
    priority: "MANDATORY"
  },
  {
    name: "Travel Rule Message Broker",
    detail: "IVMS101 payload transmitted peer-to-peer via protocol (Notabene,
    action: "Travel Rule protocol",
    priority: "HIGH"
  },
  {
    name: "Case Management System",
    detail: "All alerts flow into a case queue (Actimize, Quantexa, or open-s
    action: "Case workflow tool",
    priority: "HIGH"
  }
],
ai: [
  {
    name: "Behavioral Anomaly Detection",
    detail: "Unsupervised ML (autoencoders, isolation forests) models each cu
    tech: "Unsupervised ML",
    model: "Autoencoder / Isolation Forest"
  },
  {
    name: "Graph Neural Network (GNN) Analysis",
    detail: "Maps transaction networks as graphs. Detects layering rings, cir
    tech: "Graph ML",
    model: "GraphSAGE / Neo4j GDS"
  },
  {
    name: "SAR Narrative Generator",
    detail: "When an analyst decides to file a SAR, the AI drafts the FinCEN-
    tech: "LLM + structured prompting",
    model: "Claude / GPT-4o"
  },
  {
    name: "False Positive Reducer",
    detail: "Supervised classifier trained on historical analyst decisions (f
    tech: "Supervised Classification",
    model: "LightGBM / Random Forest"
  }
]

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},
{
  id: "blockchain",
  code: "L4",
  label: "On-Chain Intelligence",
  color: "#4BC84B",
  icon: "🔗",
  subtitle: "Digital Asset-Specific Controls",
  description: "Traditional AML tools miss the on-chain dimension. This layer i
nonAI: [
  {
    name: "Multi-Chain Address Labeling",
    detail: "Maintain or subscribe to a labeled address database covering Bit
    action: "Database subscription",
    priority: "HIGH"
  },
  {
    name: "DeFi Protocol Risk Classification",
    detail: "Not all DeFi is equal. Classify protocols by KYC status, smart c
    action: "Protocol registry",
    priority: "MEDIUM"
  },
  {
    name: "NFT & Token Layering Controls",
    detail: "NFT wash-trading and rapid token swaps are used for layering. Mo
    action: "Pattern rules",
    priority: "MEDIUM"
  },
  {
    name: "Hot/Cold Wallet Segregation",
    detail: "Operational wallets (hot) must be segregated from customer fund
    action: "Wallet governance SOP",
    priority: "HIGH"
  }
],
ai: [
  {
    name: "Mixer / Tumbler Detection",
    detail: "ML classifier identifies mixer-usage patterns: CoinJoin clusters
    tech: "Graph Classification",
    model: "Custom GNN"
  },
  {
    name: "Smart Contract Risk Analyzer",
    detail: "Static analysis + LLM audit of smart contract code. Flags rug-pu
    tech: "Static Analysis + LLM",
    model: "Slither + Claude"
  }
]

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    },
    {
      name: "Cross-Chain Tracing",
      detail: "Bridges break the transaction trail. AI agents follow funds across chains.",
      tech: "Multi-chain graph ML",
      model: "Custom pipeline"
    }
  ]
},
{
  id: "reporting",
  code: "L5",
  label: "Regulatory Reporting & Filing",
  color: "#9B4BC8",
  icon: "📁",
  subtitle: "SAR, CTR & Audit Automation",
  description: "Getting the data right is step one. Filing it correctly, on time, is step two.",
  nonAI: [
    {
      name: "BSA E-Filing System Integration",
      detail: "Direct API or batch submission to FinCEN's BSA E-Filing portal.",
      action: "API integration",
      priority: "MANDATORY"
    },
    {
      name: "SAR Decision Audit Trail",
      detail: "Document every alert: who reviewed it, what was found, why filed.",
      action: "Case documentation SOP",
      priority: "MANDATORY"
    },
    {
      name: "Record Retention System",
      detail: "BSA requires 5-year retention for all transaction records, KYC documents.",
      action: "Immutable storage setup",
      priority: "MANDATORY"
    },
    {
      name: "Examiner Readiness Package",
      detail: "Pre-built report set for FinCEN/state examinations: risk assessment, SAR analysis.",
      action: "Report templates",
      priority: "HIGH"
    }
  ],
  ai: [
    {
      name: "Auto-SAR Drafting Engine",
      detail: "Pulls alert data, customer profile, transaction timeline, and generates SAR drafts."
    }
  ]
}

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    tech: "LLM + template engine",
    model: "Claude Sonnet"
  },
  {
    name: "CTR Auto-Population",
    detail: "When cash transaction ≥ $10,000 is detected, CTR form fields are",
    tech: "Form automation",
    model: "Rules + LLM QA"
  },
  {
    name: "Compliance Dashboard Intelligence",
    detail: "Real-time KPIs: alert volume, SAR filing rate, false positive ra",
    tech: "BI + Anomaly Detection",
    model: "Custom analytics"
  }
]
},
{
  id: "governance",
  code: "L6",
  label: "Governance & Continuous Improvement",
  color: "#C87A4B",
  icon: "🏛️",
  subtitle: "The Oversight Layer",
  description: "The system must watch itself. Independent audit, model validati",
  nonAI: [
    {
      name: "Independent Annual AML Audit",
      detail: "FinCEN requires independent testing of the entire AML program –",
      action: "Annual engagement",
      priority: "MANDATORY"
    },
    {
      name: "BSA Officer Accountability",
      detail: "Designated BSA Officer must have direct board access, authority",
      action: "Role charter",
      priority: "MANDATORY"
    },
    {
      name: "Staff AML Training Program",
      detail: "Annual training for all employees; role-specific modules for com",
      action: "LMS integration",
      priority: "MANDATORY"
    },
    {
      name: "Risk Assessment (Annual)",
      detail: "Formal written enterprise-wide risk assessment covering: custome

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        action: "Risk matrix update",
        priority: "MANDATORY"
    }
],
ai: [
    {
        name: "Model Risk Management (MRM)",
        detail: "All AI models used in compliance must be validated, documented,
        tech: "MLOps pipeline",
        model: "MLflow / Weights & Biases"
    },
    {
        name: "AI Training Simulations",
        detail: "Generate synthetic red-flag scenarios and new typology cases. Us
        tech: "Synthetic data + LLM",
        model: "Claude + scenario engine"
    },
    {
        name: "Feedback Loop Engine",
        detail: "Every analyst decision (file/close) flows back as labeled traini
        tech: "Active Learning ML",
        model: "Custom pipeline"
    }
]
}
];

```

```

const techStack = [
    { category: "Blockchain Analytics", tools: ["Chainalysis", "Elliptic", "TRM Lab
    { category: "Identity / KYC", tools: ["Persona", "Onfido", "Jumio", "Stripe Ide
    { category: "Sanctions & PEP Screening", tools: ["Refinitiv World-Check", "Dow
    { category: "Travel Rule Protocol", tools: ["Notabene", "Sygna Bridge", "Verify
    { category: "Case Management", tools: ["Actimize", "Quantexa", "NICE Actimize",
    { category: "AI / ML Infrastructure", tools: ["AWS SageMaker", "Vertex AI", "Az
    { category: "Graph Database", tools: ["Neo4j", "Amazon Neptune", "TigerGraph"],
    { category: "Data Storage & Retention", tools: ["AWS S3 Object Lock", "Azure Im
    { category: "Regulatory Filing", tools: ["FinCEN BSA E-Filing", "NMLS", "State
    { category: "Orchestration & Monitoring", tools: ["Apache Kafka", "Airflow", "D
];

```

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const priorities = { MANDATORY: "#C84B4B", HIGH: "#C8A84B", MEDIUM: "#4B8FC8" };

```

```

export default function AMLFramework() {
    const [activeLayer, setActiveLayer] = useState("regulatory");
    const [activeTab, setActiveTab] = useState("nonAI");
    const [stackFilter, setStackFilter] = useState("all");

```

```

const layer = layers.find(l => l.id === activeLayer);
const filteredStack = stackFilter === "all" ? techStack : techStack.filter(t =>

return (
  <div style={{
    fontFamily: "'DM Mono', 'Courier New', monospace",
    background: "#0A0C10",
    color: "#C9D1D9",
    minHeight: "100vh",
    padding: "0",
  }}>
    <style>{`
      @import url('https://fonts.googleapis.com/css2?family=DM+Mono:wght@300;400;500;700;900&display=block');

      * { box-sizing: border-box; margin: 0; padding: 0; }

      .header-stripe {
        background: linear-gradient(90deg, #0D1117 0%, #161B22 50%, #0D1117 100%);
        border-bottom: 1px solid #30363D;
        padding: 24px 32px;
      }

      .badge {
        display: inline-block;
        padding: 3px 8px;
        border-radius: 3px;
        font-size: 9px;
        font-weight: 500;
        letter-spacing: 1.5px;
        text-transform: uppercase;
      }

      .layer-nav {
        display: flex;
        gap: 0;
        overflow-x: auto;
        border-bottom: 1px solid #30363D;
        background: #0D1117;
        padding: 0 24px;
      }

      .layer-btn {
        background: none;
        border: none;
        border-bottom: 2px solid transparent;
        color: #8B949E;
        padding: 14px 16px;
      }
    `}
  </style>

```

```

    cursor: pointer;
    font-family: inherit;
    font-size: 11px;
    letter-spacing: 0.5px;
    white-space: nowrap;
    transition: all 0.2s;
    display: flex;
    align-items: center;
    gap: 6px;
}

.layer-btn:hover { color: #E6EDF3; }
.layer-btn.active { color: #E6EDF3; border-bottom-color: var(--layer-colo

.tab-btn {
    background: none;
    border: 1px solid #30363D;
    color: #8B949E;
    padding: 8px 20px;
    cursor: pointer;
    font-family: inherit;
    font-size: 11px;
    letter-spacing: 1px;
    text-transform: uppercase;
    transition: all 0.2s;
}
.tab-btn:first-child { border-radius: 4px 0 0 4px; }
.tab-btn:last-child { border-radius: 0 4px 4px 0; }
.tab-btn.active { background: #161B22; color: #E6EDF3; border-color: #58A

.card {
    background: #0D1117;
    border: 1px solid #21262D;
    border-radius: 6px;
    padding: 18px;
    margin-bottom: 12px;
    transition: border-color 0.2s;
}
.card:hover { border-color: #30363D; }

.pill {
    display: inline-block;
    background: #161B22;
    border: 1px solid #30363D;
    border-radius: 20px;
    padding: 4px 12px;
    font-size: 10px;

```

```

    color: #8B949E;
    margin: 2px;
}

.scrollbar-thin::-webkit-scrollbar { height: 4px; }
.scrollbar-thin::-webkit-scrollbar-track { background: #0A0C10; }
.scrollbar-thin::-webkit-scrollbar-thumb { background: #30363D; border-ra

.flow-step {
  display: flex;
  align-items: flex-start;
  gap: 16px;
  padding: 14px 0;
  border-bottom: 1px solid #21262D;
}
.flow-step:last-child { border-bottom: none; }

.stack-chip {
  background: #161B22;
  border: 1px solid #30363D;
  border-radius: 4px;
  padding: 6px 12px;
  font-size: 11px;
  color: #C9D1D9;
  white-space: nowrap;
}
`}</style>

{/* Header */}
<div className="header-stripe">
  <div style={{ display: "flex", alignItems: "flex-start", justifyContent:
    <div>
      <div style={{ display: "flex", alignItems: "center", gap: 10, marginB
        <span style={{ fontSize: 20 }}>■</span>
        <span style={{ fontFamily: "'Space Grotesk', sans-serif", fontSize:
      </div>
      <h1 style={{ fontFamily: "'Space Grotesk', sans-serif", fontSize: 22,
        AML/BSA Compliance Automation Framework
      </h1>
      <p style={{ color: "#8B949E", fontSize: 12, marginTop: 6, letterSpacing:
        Digital Assets & VASP · FCF Global Transactions · FinCEN | OFAC | F
      </p>
    </div>
    <div style={{ display: "flex", gap: 6, flexWrap: "wrap" }}>
      [{"FinCEN", "OFAC", "FATF", "BSA", "KYC/AML", "Travel Rule"].map(t =>
        <span key={t} className="badge" style={{ background: "#161B22", bor
      )})

```

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    </div>
  </div>

  {/ * Program status bar */}
  <div style={{ display: "flex", gap: 24, marginTop: 18, paddingTop: 16, bo
    {
      { label: "Framework Layers", value: "6" },
      { label: "Control Categories", value: "40+" },
      { label: "Mandatory Controls", value: "18" },
      { label: "AI/ML Components", value: "16" },
      { label: "Regulatory Bodies", value: "FinCEN · OFAC · FATF · State" }
    ].map(s => (
      <div key={s.label}>
        <div style={{ fontSize: 18, fontWeight: 500, color: "#C8A84B", font
          <div style={{ fontSize: 10, color: "#8B949E", letterSpacing: 0.5 }}
        </div>
      </div>
    )))
  </div>
</div>

{/ * Layer Navigation */}
<div className="layer-nav scrollbar-thin">
  {layers.map(l => (
    <button
      key={l.id}
      className={`layer-btn ${activeLayer === l.id ? "active" : ""}`}
      style={{ "--layer-color": l.color }}
      onClick={() => setActiveLayer(l.id)}
    >
      <span>{l.icon}</span>
      <span style={{ fontFamily: "'Space Grotesk', sans-serif", fontWeight:
        {l.code} · {l.label}
      </span>
    </button>
  )))
</div>

{/ * Main Content */}
<div style={{ padding: "24px 32px", maxWidth: 1100 }}>

  {/ * Layer Header */}
  <div style={{ marginBottom: 24, paddingBottom: 20, borderBottom: "1px sol
    <div style={{ display: "flex", alignItems: "center", gap: 12, marginBot
      <span style={{ fontSize: 28 }}>{layer.icon}</span>
    </div>
    <div style={{ display: "flex", alignItems: "center", gap: 10 }}>
      <h2 style={{ fontFamily: "'Space Grotesk', sans-serif", fontSize:

```

```

        <span className="badge" style={{ background: layer.color + "22",
        </div>
        <p style={{ color: layer.color, fontSize: 12, marginTop: 2, lettersS
        </div>
        </div>
        <p style={{ color: "#8B949E", fontSize: 13, lineHeight: 1.6, maxWidth:
        </div>

    { /* Tabs */}
    <div style={{ display: "flex", gap: 0, marginBottom: 20 }}>
        <button className={`tab-btn ${activeTab === "nonAI" ? "active" : ""}`}
            ⚙ Non-AI Controls ({layer.nonAI.length})
        </button>
        <button className={`tab-btn ${activeTab === "ai" ? "active" : ""}`} onClick
            💎 AI/ML Layer ({layer.ai.length})
        </button>
    </div>

    { /* Non-AI Controls */}
    {activeTab === "nonAI" && (
        <div>
            {layer.nonAI.map((item, i) => (
                <div key={i} className="card">
                    <div style={{ display: "flex", alignItems: "flex-start", justifyC
                        <div style={{ display: "flex", alignItems: "center", gap: 10 }}
                            <span style={{ color: "#58A6FF", fontSize: 11, fontWeight: 50
                                {String(i + 1).padStart(2, "0")}
                            </span>
                            <h3 style={{ fontFamily: "'Space Grotesk', sans-serif", fontS
                        </div>
                        <div style={{ display: "flex", gap: 8, alignItems: "center" }}>
                            <span className="badge" style={{ background: priorities[item.
                                {item.priority}
                            </span>
                            <span className="pill">{item.action}</span>
                        </div>
                    </div>
                <p style={{ color: "#8B949E", fontSize: 12, lineHeight: 1.7, padd
            </div>
        )})
    </div>
    )}

    { /* AI/ML Layer */}
    {activeTab === "ai" && (
        <div>
            <div style={{ background: "#161B22", border: "1px solid #30363D", bor

```

```

    ▲ All AI/ML models used in AML decisions require <strong style={{ c
</div>
{layer.ai.map((item, i) => (
  <div key={i} className="card" style={{ borderLeft: `3px solid ${lay
    <div style={{ display: "flex", alignItems: "flex-start", justifyC
      <div style={{ display: "flex", alignItems: "center", gap: 10 }}
        <span style={{ color: layer.color, fontSize: 11, fontWeight:
          <h3 style={{ fontFamily: "'Space Grotesk', sans-serif", fontS
        </div>
      <div style={{ display: "flex", gap: 6 }}>
        <span className="pill">⚡ {item.tech}</span>
        <span className="pill">🤖 {item.model}</span>
      </div>
    </div>
    <p style={{ color: "#8B949E", fontSize: 12, lineHeight: 1.7, padd
  </div>
  )))
</div>
))

{/* Technology Stack Section */}
<div style={{ marginTop: 36, paddingTop: 28, borderTop: "1px solid #21262
  <div style={{ display: "flex", alignItems: "center", justifyContent: "s
    <h3 style={{ fontFamily: "'Space Grotesk', sans-serif", fontSize: 16,
      🛠 Recommended Technology Stack
    </h3>
    <div style={{ display: "flex", gap: 0 }}>
      {[["all", "All"], ["vendor", "SaaS Vendors"], ["infra", "Infrastruc
        <button
          key={v}
          className={`tab-btn ${stackFilter === v ? "active" : ""}`}
          style={{ fontSize: 10, padding: "6px 12px" }}
          onClick={() => setStackFilter(v)}
        >{l}</button>
      </div>
    </div>
  </div>
  <div style={{ display: "grid", gridTemplateColumns: "repeat(auto-fill,
    {filteredStack.map((cat, i) => (
      <div key={i} className="card" style={{ margin: 0 }}>
        <div style={{ fontSize: 10, color: "#8B949E", letterSpacing: 1, t
          <div style={{ display: "flex", flexWrap: "wrap", gap: 6 }}>
            {cat.tools.map(t => (
              <span key={t} className="stack-chip">{t}</span>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>

```

```

    )))
  </div>
</div>

{/* Workflow Orchestration */}
<div style={{ marginTop: 36, paddingTop: 28, borderTop: "1px solid #21262"
  <h3 style={{ fontFamily: "'Space Grotesk', sans-serif", fontSize: 16, f
    ⚡ Transaction Compliance Workflow – End to End
  </h3>
  <p style={{ color: "#8B949E", fontSize: 12, marginBottom: 20 }}>How a s

  [[
    { step: "01", phase: "PRE-ONBOARDING", label: "Customer hits sign-up"
    { step: "02", phase: "ONBOARDING GATE", label: "Identity verified & r
    { step: "03", phase: "TRANSACTION INITIATION", label: "Customer submi
    { step: "04", phase: "PRE-SETTLEMENT GATE", label: "Hard block or pas
    { step: "05", phase: "SETTLEMENT", label: "Transaction clears", actio
    { step: "06", phase: "POST-TRANSACTION MONITORING", label: "Behaviora
    { step: "07", phase: "ALERT & CASE MANAGEMENT", label: "Alert trigger
    { step: "08", phase: "REGULATORY FILING", label: "SAR/CTR auto-drafte
  ].map((step) => (
    <div key={step.step} className="flow-step">
      <div style={{ minWidth: 36, height: 36, borderRadius: "50%", backgr
        {step.step}
      </div>
      <div style={{ flex: 1 }}>
        <div style={{ display: "flex", alignItems: "center", gap: 8, marg
          <span className="badge" style={{ background: step.color + "22",
            <span style={{ fontFamily: "'Space Grotesk', sans-serif", fontS
          </div>
          <p style={{ color: "#8B949E", fontSize: 12, lineHeight: 1.6 }}>{s
        </div>
      </div>
    )))
  </div>

{/* Footer */}
<div style={{ marginTop: 40, paddingTop: 20, borderTop: "1px solid #21262
  <div style={{ fontSize: 10, color: "#484F58", letterSpacing: 0.5 }}>
    FRAMEWORK VERSION 1.0 · FCF GLOBAL · AML/BSA DIGITAL ASSETS
  </div>
  <div style={{ fontSize: 10, color: "#484F58" }}>
    Consult qualified BSA/AML counsel before implementation · This framew
  </div>
</div>
</div>
</div>

```

);
}